HUJI Applicative Research Report — Healthcare — Week 36, 2026

7 high-potential paper(s) · Healthcare · Weekly · Week 36, 2026 · Generated August 31, 2026

Hebrew University researchers are advancing cutting-edge solutions across pivotal sectors, from transformative medical treatments to sustainable agriculture and fundamental scientific platforms. Highlights include a novel stem cell therapy poised to repair myelin in neurological disorders and a groundbreaking genetic delivery system overcoming a key hurdle in gene therapies. Additionally, innovations in 'smart' gene therapy for epilepsy, pest-resistant crops for global food security, and enhanced drug combinations for cystic fibrosis demonstrate broad impact and commercial readiness.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Tamir Ben-Hur** — Neuroscience, Immunology, Clinical, Synthetic Biology
- **Eylon Yavin** — Drug Discovery, Genomics, Synthetic Biology
- **Tawfeeq Shekh-Ahmad** — Drug Discovery, Neuroscience, Genomics, Clinical
- **Osnat Malka** — AgriTech, Genomics, Synthetic Biology
- **Batsheva Kerem** — Drug Discovery, Genomics, Clinical
- **Dmitry Tsvelikhovsky** — Drug Discovery, Materials, Synthetic Biology
- **Ayelet N Landau** — Diagnostics, Medical Device, Software/AI, Neuroscience

HEALTHCARE — RESEARCH HIGHLIGHTS

1. Pro-repair properties of a human embryonic stem cell-derived astrocyte cell therapy in demyelinating disorders. 36/50

"Off-the-shelf stem cell therapy offers a new paradigm for remyelination in MS and other neurological diseases."

Main researcher: Tamir Ben-Hur · tamir@hadassah.org.il.

Faculty of Medicine, Hebrew University of Jerusalem, Jerusalem, Israel; The Department of Neurology, The Agnes-Ginges Center for Human Neurogenetics, Hadassah-Hebrew University Medical Center, Jerusalem, Israel. Electronic address: tamir@hadassah.org.il.

Published in Stem cell reports · 2026-06

ABOUT THIS RESEARCH

Researchers developed a clinical-grade astrocyte therapy derived from embryonic stem cells that helps the brain repair damaged myelin sheaths. The therapy works by clearing debris and creating a supportive environment that encourages the growth of new insulating layers for nerves.

COMMERCIAL ANGLE

Development of a scalable, off-the-shelf cell therapy for remyelination in Multiple Sclerosis and other demyelinating neurological disorders.

COMMERCIALISATION METRICS

Novelty 6/10

While the use of hESC-derived astrocytes is a sophisticated approach to remyelination, the 'bystander mechanism' and the focus on microglial debris clearance are established concepts in neuroregeneration, making this more of an incremental application of cell therapy than a groundbreaking scientific paradigm shift.

Commercial Potential 8/10

The therapy uses clinical-grade, scalable hESC-derived cells targeting a high-unmet-need market (demyelinating disorders) with demonstrated safety and a clear mechanism of action. The 'clinical-grade' status and scalability suggest it is already positioned for translational development rather than being purely theoretical.

Market Size 9/10

The therapy targets demyelinating disorders (such as Multiple Sclerosis), which represent a massive, high-value patient population with significant unmet medical needs. The mechanism's ability to address multiple facets of remyelination failure suggests a broad, multi-indication platform potential within the neurodegenerative market.

Tech Readiness 5/10

The technology has transitioned from basic lab discovery to 'clinical-grade' manufacturing and has demonstrated efficacy in animal models (lysolecithin-induced), but it remains in the preclinical validation stage rather than undergoing human clinical trials.

IP Strength 8/10

The development of a 'clinical-grade' hESC-derived astrocyte product (AstroRx) suggests a proprietary, scalable manufacturing process and a specific cell composition that is highly patentable. The multi-modal mechanism of action—addressing debris clearance and OPC differentiation—provides multiple layers of therapeutic defensibility and platform potential for various CNS disorders.

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2. Multiple Charged Purine (MCP) PNA as a Simple Mode for Cellular Uptake. 33/50

"Novel synthetic DNA analogue enables natural cellular uptake, revolutionizing genetic therapy delivery."

Main researcher: Eylon Yavin

The Institute for Drug Research, The School of Pharmacy, The Faculty of Medicine, The Hebrew University of Jerusalem, Hadassah Ein-Kerem, Jerusalem 9112102, Israel.

Published in ACS polymers Au · 2026-06

ABOUT THIS RESEARCH

Researchers developed a new type of synthetic DNA analogue called MCP PNA that can naturally enter cells without needing complex delivery vehicles. This solves a major historical bottleneck where these stable molecules were too large or poorly charged to cross cell membranes.

COMMERCIAL ANGLE

Development of a versatile, high-stability platform for intracellular genetic targeting and antisense therapeutics without the need for expensive lipid nanoparticles or viral vectors.

COMMERCIALISATION METRICS

Novelty 6/10

The work provides a practical chemical modification to improve PNA uptake, but utilizing charge modification for cellular delivery is a known strategy in oligonucleotide chemistry rather than a groundbreaking platform shift.

Commercial Potential 8/10

This represents a platform-enabling technology that solves a fundamental bottleneck (cellular uptake) for PNA, significantly expanding the addressable market for PNA-based therapeutics and diagnostics.

Market Size 9/10

The work addresses a fundamental delivery bottleneck for PNA, a platform with massive potential across antisense therapy, diagnostics, and gene editing markets. Solving cellular uptake enables a wide range of therapeutic applications, making this a platform-enabling advancement with a very large total addressable market.

Tech Readiness 3/10

The work focuses on fundamental chemical modification of PNA for cellular uptake, representing basic proof-of-concept laboratory research without evidence of in vivo efficacy or a developed delivery platform.

IP Strength 7/10

The introduction of a specific chemical modification (MCP) to solve a fundamental PNA delivery bottleneck is highly patentable as a composition of matter. While it enables a platform for cellular uptake, its ultimate strength depends on the breadth of the claims regarding the purine modifications.

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3. Activity-dependent antioxidant gene therapy protects against epileptogenesis. 31/50

""Smart" gene therapy prevents epilepsy development by activating antioxidants only when needed."

Main researcher: Tawfeeq Shekh-Ahmad · Tawfeeq.Shekh-Ahmad@mail.huji.ac.il.

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Published in Signal transduction and targeted therapy · 2026-06

ABOUT THIS RESEARCH

This research describes a gene therapy approach that uses natural brain activity to trigger the production of antioxidant enzymes. This mechanism aims to prevent the development of epilepsy following an initial brain injury.

COMMERCIAL ANGLE

Development of smart, responsive gene therapies for neurological disorders that activate only during specific physiological states to minimize side effects.

COMMERCIALISATION METRICS

Novelty 7/10

The work is highly innovative for shifting the therapeutic paradigm from reactive seizure control to proactive, activity-dependent gene modulation to prevent epileptogenesis. However, while the mechanism is sophisticated, antioxidant gene therapy is an established concept, making the novelty incremental rather than a complete disruption of biological principles.

Commercial Potential 7/10

The technology targets a high-unmet-need clinical indication (epileptogenesis) with a specific mechanism (antioxidant gene therapy), offering a clear path toward a specialized therapeutic product. However, the lack of abstract details regarding delivery vectors and scalability limits a higher score.

Market Size 7/10

The target market for epilepsy therapeutics is massive and high-value, representing a multi-billion dollar global neurology segment. However, the score is tempered by the specific niche of epileptogenesis prevention, which is a subset of the broader epilepsy market and faces complex clinical adoption hurdles.

Tech Readiness 3/10

The title suggests a fundamental mechanistic study using gene therapy in a disease model, which typically implies preclinical, lab-based proof-of-concept research rather than a developed therapeutic candidate.

IP Strength 7/10

The use of activity-dependent promoters for targeted gene therapy provides a strong basis for composition-of-matter and method-of-use patents, though defensibility depends on the novelty of the specific promoter/transgene combination.

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4. Creating resistance to the whitefly <i>Bemisia tabaci</i> in cassava through RNAi-mediated targeting of multiple insect metabolic processes. 31/50

"RNAi-engineered cassava offers robust whitefly resistance, safeguarding critical global food supplies."

Main researcher: Osnat Malka

Published · 2026-05

ABOUT THIS RESEARCH

Researchers developed a way to make cassava plants resistant to whiteflies, which are the primary carriers of devastating diseases in East Africa. They used RNA interference (RNAi) to target and disrupt essential metabolic processes within the insects.

COMMERCIAL ANGLE

Development of genetically engineered, pest-resistant crop seeds and RNAi-based biopesticides to reduce yield loss in global cassava production.

COMMERCIALISATION METRICS

Novelty 6/10

While targeting multiple metabolic pathways via RNAi is a robust strategy, host-induced gene silencing (HIGS) for pest control is a well-established technique; the novelty here lies primarily in the application to a specific crop-pest system rather than a fundamental scientific breakthrough.

Commercial Potential 7/10

The technology addresses a massive, billion-dollar agricultural pain point with a clear product path (transgenic resistant cultivars). However, the commercial potential is slightly tempered by the regulatory hurdles of GMOs in African markets and the necessity for multi-gene targeting to prevent insect resistance.

Market Size 8/10

The target market is vast and economically critical, with annual losses exceeding $1.25 billion and direct impact on food security for millions in East Africa. The RNAi approach offers a scalable, platform-like solution to a 'super-abundant' pest affecting a primary global staple crop.

Tech Readiness 3/10

The research is in the early laboratory stage, focusing on the identification of candidate genes and the production of RNAi constructs, which precedes field testing and regulatory approval.

IP Strength 7/10

The use of multi-gene RNAi targeting specific metabolic pathways allows for strong composition-of-matter patents on the constructs and potential 'stacking' strategies to prevent resistance. However, defensibility is slightly tempered by the general nature of RNAi techniques and potential regulatory hurdles in agricultural biotech.

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5. SPL84 and trikafta® exhibit comparable and additive effects in patient-derived HBE cells carrying the 3849+10kb C→T/F508del CFTR variants. 31/50

"New combination therapy enhances CFTR function for cystic fibrosis patients with specific mutations."

Main researcher: Batsheva Kerem · batshevak@savion.huji.ac.il.

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Published in Journal of cystic fibrosis : official journal of the European Cystic Fibrosis Society · 2026-06

ABOUT THIS RESEARCH

Researchers found that combining a new inhaled antisense oligonucleotide (SPL84) with the existing drug Trikafta® provides additive benefits for patients with a specific cystic fibrosis mutation. This combination therapy effectively restores CFTR protein function in human lung cells, potentially offering better clinical outcomes than current standards of care alone.

COMMERCIAL ANGLE

Developing a combination therapy or a precision medicine regimen that pairs existing CFTR modulators with targeted antisense oligonucleotides to capture the underserved segment of patients with modest responses to current treatments.

COMMERCIALISATION METRICS

Novelty 6/10

The research is an incremental optimization study focused on combination therapy rather than a fundamental discovery of a new biological mechanism or platform. While the additive effect of an ASO with an existing standard of care is clinically relevant, it represents an evolutionary application of known splicing modulation and CFTR corrector technologies.

Commercial Potential 8/10

The study demonstrates a clear additive therapeutic strategy for a specific, high-value patient population, providing a strong de-risking signal for a combination product approach. The focus on an inhaled ASO targeting a known clinical gap in current standard-of-care (Trikafta) represents a highly viable and direct path to clinical application.

Market Size 4/10

The target market is limited to a highly specific genetic sub-population (3849+10kb variant) within the already established cystic fibrosis patient base. While clinically valuable for non-responders, the addressable demand is niche and constrained by the prevalence of this specific splicing mutation.

Tech Readiness 6/10

The asset has already completed a Phase 2a clinical trial, indicating significant de-risking; however, the current paper focuses on ex vivo validation in patient-derived cells rather than advancing the clinical trial phase.

IP Strength 7/10

The innovation utilizes a specific antisense oligonucleotide (ASO) sequence targeting a known splicing defect, which is highly patentable and provides a clear pathway for composition-of-matter protection. However, the defensibility is slightly tempered by the fact that it is a combination therapy with an existing standard-of-care, which may face competition from other splicing modulators.

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6. Proline-promoted electrophilic dichloromethylation of α-enaminones via stereochemically gated resolution. **30/50**

"Novel metal-free synthesis method streamlines production of complex pharmaceutical and material building blocks."

Main researcher: Dmitry Tsvelikhovsky · dmitryt@ekmd.huji.ac.il.

The Institute for Drug Research, Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, 9112102, Israel. dmitryt@ekmd.huji.ac.il.

Published in Nature communications · 2026-04

ABOUT THIS RESEARCH

Researchers have developed a new metal-free method to attach polyhalomethyl groups to specific organic molecules using proline as a catalyst. This process is highly controllable and can be used to build complex chemical structures or create aromatic rings through a programmed cascade reaction.

COMMERCIAL ANGLE

This platform provides a more efficient and sustainable way to synthesize complex building blocks used in pharmaceutical manufacturing and advanced materials science.

COMMERCIALISATION METRICS

Novelty **8/10**

The research introduces a novel closed-shell, metal-free mechanism for polyhalomethylation that shifts away from traditional radical/carbene pathways. The discovery of topology-driven divergence—where ring size dictates either functionalization or spontaneous cascade aromatization—represents a significant, programmable platform for synthetic methodology.

Commercial Potential **6/10**

The paper describes a highly versatile, metal-free synthetic platform that could streamline the production of complex building blocks for medicinal chemistry. However, its commercial value is currently limited to a specialized tool for process chemists rather than a stand-alone product or a transformative platform with broad industrial scalability.

Market Size **6/10**

The platform offers valuable versatility for medicinal chemistry and fine chemical synthesis by enabling novel functionalization of building blocks, but its market size is constrained to the niche specialty chemical and R&D reagent sectors rather than a large-scale therapeutic or industrial commodity market.

Tech Readiness **3/10**

This is fundamental discovery research focused on mechanistic understanding and new synthetic methodology at the academic lab stage. While it demonstrates a novel chemical platform, it lacks the process optimization, scalability testing, or applied proof-of-concept required for higher TRL scores.

IP Strength **7/10**

The discovery of a unique, metal-free, closed-shell mechanistic class offers strong potential for broad composition-of-matter and process patents in medicinal chemistry. However, the utility is currently limited to fundamental synthetic methodology, which may face challenges in demonstrating high-value industrial exclusivity compared to targeted therapeutic or material platforms.

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7. Universal rhythmic architecture uncovers two modes of neural dynamics. 28/50

"Universal brain activity patterns enable next-generation neurodiagnostic software and advanced BCIs."

Main researcher: Ayelet N Landau

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Published in Nature communications · 2026-05

ABOUT THIS RESEARCH

Researchers have discovered a universal pattern in brain activity that distinguishes between sustained rhythms and brief, transient bursts. This framework allows for a more nuanced understanding of how different brain regions process ongoing information versus responding to new environmental stimuli.

COMMERCIAL ANGLE

This universal architecture can be used to develop highly personalized neurodiagnostic software and advanced brain-computer interfaces that distinguish between stable cognitive states and rapid sensory responses.

COMMERCIALISATION METRICS

Novelty 8/10

The research shifts the fundamental paradigm of neural analysis from a frequency-only model to a two-dimensional architecture (frequency + rhythmicity), providing a potentially transformative framework for neurotechnological diagnostics. This provides a platform-enabling discovery that unifies disparate datasets across species and clinical states into a single organizing principle.

Commercial Potential 5/10

While the framework offers high potential for improving the precision of neurodiagnostic biomarkers and personalized EEG analysis, it remains a fundamental descriptive discovery rather than a specific device or therapeutic intervention. Its commercial value depends entirely on its integration into future clinical software or neurotechnology platforms.

Market Size 8/10

The findings provide a platform-enabling framework for universal neurophysiological analysis, addressing a massive market spanning clinical diagnostics, neuroprosthetics, and brain-computer interfaces across diverse patient populations.

Tech Readiness 3/10

The paper presents fundamental discovery-based neuroscience that provides a new theoretical framework for understanding neural dynamics. While it offers potential for future diagnostic biomarkers, it is currently at a basic research stage without a defined prototype, hardware integration, or clinical validation pipeline.

IP Strength 4/10

The work describes a scientific framework and a descriptive method for analyzing existing neural data rather than a novel hardware device or a proprietary biochemical entity. While the 'rhythmicity measure' could potentially be patented as a diagnostic algorithm, the primary output is foundational knowledge that is difficult to defend against competitors using alternative mathematical modeling.

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